Tutorial-RK_Butcher_Table_Dictionary

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-1 Explicit Runge Kutta methods and their Butcher tables

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-1.2 This tutorial notebook stores known explicit Runge Kutta-like methods as Butcher tables in a Python dictionary format.

Notebook Status: Validated

Validation Notes: This tutorial notebook has been confirmed to be **self-consistent with its corresponding NRPy+ module**, as documented below. In addition, each of these Butcher tables has been verified to yield an RK method to the expected local truncation error in a challenging battery of ODE tests in the RK Butcher Table Validation tutorial notebook.

-1.2.1 NRPy+ Source Code for this module: MoLtimestepping/RK_Butcher_Table_Dictionary.py

-1.3 Introduction:

The family of explicit Runge Kutta-like methods are commonly used when numerically solving ordinary differential equation (ODE) initial value problems of the form

$$y'(t) = f(y,t), \quad y(t_0) = y_0.$$

These methods can be extended to solve time-dependent partial differential equations (PDEs) via the Method of Lines. In the Method of Lines, the above ODE can be generalized to N coupled ODEs, all written as first-order-in-time PDEs of the form

$$\partial_t \mathbf{u}(t,x,y,u_1,u_2,u_3,...) = \mathbf{f}(t,x,y,...,u_1,u_{1,x},...),$$

where $\mathbf{u}$ and $\mathbf{f}$ are vectors. The spatial partial derivatives of components of $\mathbf{u}$, e.g., $u_{1,x}$, may be computed using approximate numerical differentiation, like finite differences.

As any explicit Runge-Kutta method has its own unique local truncation error, can in principle be used to solve time-dependent PDEs using the Method of Lines, and may be stable under different Courant-Friedrichs-Lewy (CFL) conditions, it is useful to have multiple methods at one's disposal. **This module provides a number of such methods.**

More details about the Method of Lines is discussed further in the Tutorial-RK_Butcher_Table_Generating_C_Code module where we generate the C code to implement the Method of Lines, and additional description can be found in the Numerically Solving the Scalar Wave Equation: A Complete C Code NRPy+ tutorial notebook.

0 Table of Contents

This notebook is organized as follows

1. Step 1: Initialize needed Python modules
2. Step 2: The Family of Explicit Runge-Kutta-Like Schemes (Butcher Tables)
    1. Step 2a: Generating a Dictionary of Butcher Tables for Explicit Runge Kutta Techniques
        1. Step 2.a.i: Euler's Method
        2. Step 2.a.ii: RK2 Heun's Method
        3. Step 2.a.iii: RK2 Midpoint Method
        4. Step 2.a.iv: RK2 Ralston's Method
        5. Step 2.a.v: Kutta's Third-order Method
        6. Step 2.a.vi.: RK3 Heun's Method
        7. Step 2.a.vii: RK3 Ralston's Method
        8. Step 2.a.viii: Strong Stability Preserving Runge-Kutta (SSPRK3) Method
        9. Step 2.a.ix: Classic RK4 Method
        10. Step 2.a.x: RK5 Dormand-Prince Method
        11. Step 2.a.xi: RK5 Dormand-Prince Method Alternative
        12. Step 2.a.xii: RK5 Cash-Karp Method
        13. Step 2.a.xiii: RK6 Dormand-Prince Method
        14. Step 2.a.xiv: RK6 Luther Method
        15. Step 2.a.xv: RK8 Dormand-Prince Method
3. Step 3: Code Validation against `MoLtimestepping.RK_Butcher_Table_Dictionary` NRPy+ module
4. Step 4: The Family of Explicit Runge-Kutta-Like Schemes (Butcher Tables)
    1. Step 4a: Generating a Dictionary of Butcher Tables for Explicit Runge Kutta Techniques
        1. Step 4.a.i: Adaptive Heun-Euler Method
        2. Step 4.a.ii: Adaptibe Bogacki-Shampine Method
        3. Step 4.a.iii: Adaptibe Runge-Kutta-Fehlberg
5. Step 5: Output this notebook to LaTeX-formatted PDF file

1 Step 1: Initialize needed Python modules [Back to top]

Let's start by importing all the needed modules from Python:

[1]:
```
# Step 1: Initialize needed Python modules
import sympy as sp  # SymPy: The Python computer algebra package upon which NRPy+ depends
```

2 Step 2: The Family of Explicit Runge-Kutta-Like Schemes (Butcher Tables) [Back to top]

In general, a predictor-corrector method performs an estimate timestep from n to $n+1$, using e.g., a Runge Kutta method, to get a prediction of the solution at timestep $n+1$. This is the "predictor" step. Then it uses this prediction to perform another, "corrector" step, designed to increase the accuracy of the solution.

Let us focus on the ordinary differential equation (ODE)

$$y'(t) = f(y,t),$$

which acts as an analogue for a generic PDE $\partial_t u(t,x,y,...) = f(t,x,y,...,u,u_x,...)$.

The general family of Runge Kutta "explicit" timestepping methods are implemented using the following scheme:

$$y_{n+1} = y_n + \sum_{i=1}^{s} b_i k_i$$

where

$$k_1 = \Delta t f(y_n, t_n) \tag{1}$$
$$k_2 = \Delta t f(y_n + [a_{21}k_1], t_n + c_2\Delta t) \tag{2}$$
$$k_3 = \Delta t f(y_n + [a_{31}k_1 + a_{32}k_2], t_n + c_3\Delta t) \tag{3}$$
$$\vdots \tag{4}$$
$$k_s = \Delta t f(y_n + [a_{s1}k_1 + a_{s2}k_2 + \cdots + a_{s,s-1}k_{s-1}], t_n + c_s\Delta t). \tag{5}$$

Note s is the number of right-hand side evaluations necessary for any given method, i.e., for RK2 $s=2$ and for RK4 $s=4$, and for RK6 $s=7$. These schemes are often written in the form of a so-called "Butcher tableau" or "Butcher table."

$$\begin{array}{c|ccccc} 0 & & & & & \\ c_2 & a_{21} & & & & \\ c_3 & a_{31} & a_{32} & & & \\ \vdots & \vdots & & \ddots & & \\ c_s & a_{s_1} & a_{s2} & \cdots & a_{s,s-1} & \\ \hline & b_1 & b_2 & \cdots & b_{s-1} & b_s \end{array}$$

As an example, the "classic" fourth-order Runge Kutta (RK4) method obtains the solution $y(t)$ to the single-variable ODE $y'(t) = f(y(t),t)$ at time t_{n+1} from t_n via:

$$k_1 = \Delta t f(y_n, t_n), \tag{6}$$
$$k_2 = \Delta t f(y_n + \frac{1}{2}k_1, t_n + \frac{\Delta t}{2}), \tag{7}$$
$$k_3 = \Delta t f(y_n + \frac{1}{2}k_2, t_n + \frac{\Delta t}{2}), \tag{8}$$
$$k_4 = \Delta t f(y_n + k_3, t_n + \Delta t), \tag{9}$$
$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) + \mathcal{O}\big((\Delta t)^5\big). \tag{10}$$

Its corresponding Butcher table is constructed as follows:

0				
1/2	1/2			
1/2	0	1/2		
1	0	0	1	
	1/6	1/3	1/3	1/6

This is one example of many explicit Runge Kutta methods. Throughout the following sections we will highlight different Runge Kutta schemes and their Butcher tables from the first-order Euler's method up to and including an eighth-order method.

2.1 Step 2.a: Generating a Dictionary of Butcher Tables for Explicit Runge Kutta Techniques [Back to top]

We can store all of the Butcher tables in Python's **Dictionary** format using the curly brackets {} and 'key':value pairs. The 'key' will be the *name* of the Runge Kutta method and the value will be the Butcher table itself stored as a list of lists. The convergence order for each Runge Kutta method is also stored. We will construct the dictionary `Butcher_dict` one Butcher table at a time in the following sections.

[2]:
```
# Step 2a: Generating a Dictionary of Butcher Tables for Explicit Runge Kutta Techniques

# Initialize the dictionary Butcher_dict
Butcher_dict = {}
```

2.1.1 Step 2.a.i: Euler's Method [Back to top]

Forward Euler's method is a first order Runge Kutta method. Euler's method obtains the solution $y(t)$ at time t_{n+1} from t_n via:

$$y_{n+1} = y_n + \Delta t f(y_n, t_n)$$

with the trivial corresponding Butcher table

0	
	1

.

[3]:
```
# Step 2.a.i: Euler's Method

Butcher_dict['Euler'] = (
[[sp.sympify(0)],
["", sp.sympify(1)]]
, 1)
```

2.1.2 Step 2.a.ii: RK2 Heun's Method [Back to top]

Heun's method is a second-order RK method that obtains the solution $y(t)$ at time t_{n+1} from t_n via:

$$k_1 = \Delta t f(y_n, t_n), \tag{11}$$
$$k_2 = \Delta t f(y_n + k_1, t_n + \Delta t), \tag{12}$$
$$y_{n+1} = y_n + \frac{1}{2}(k_1 + k_2) + \mathcal{O}\big((\Delta t)^3\big) \tag{13}$$

with corresponding Butcher table

$$\begin{array}{c|cc} 0 & & \\ 1 & 1 & \\ \hline & 1/2 & 1/2 \end{array}.$$

[4]:
```
# Step 2.a.ii: RK2 Heun's Method

Butcher_dict['RK2 Heun'] = (
[[sp.sympify(0)],
[sp.sympify(1), sp.sympify(1)],
["", sp.Rational(1,2), sp.Rational(1,2)]]
, 2)
```

2.1.3 Step 2.a.iii: RK2 Midpoint Method [Back to top]

Midpoint method is a second-order RK method that obtains the solution $y(t)$ at time t_{n+1} from t_n via:

$$k_1 = \Delta t f(y_n, t_n), \tag{14}$$
$$k_2 = \Delta t f(y_n + \frac{2}{3}k_1, t_n + \frac{2}{3}\Delta t), \tag{15}$$
$$y_{n+1} = y_n + \frac{1}{2}k_2 + \mathcal{O}\big((\Delta t)^3\big) \tag{16}$$

with corresponding Butcher table

$$\begin{array}{c|cc} 0 & & \\ 1/2 & 1/2 & \\ \hline & 0 & 1 \end{array}.$$

[5]:
```
# Step 2.a.iii: RK2 Midpoint (MP) Method

Butcher_dict['RK2 MP'] = (
[[sp.sympify(0)],
[sp.Rational(1,2), sp.Rational(1,2)],
["", sp.sympify(0), sp.sympify(1)]]
, 2)
```

2.1.4 Step 2.a.iv: RK2 Ralston's Method [Back to top]

Ralston's method (see Ralston (1962), is a second-order RK method that obtains the solution $y(t)$ at time t_{n+1} from t_n via:

$$k_1 = \Delta t f(y_n, t_n), \tag{17}$$

$$k_2 = \Delta t f(y_n + \frac{1}{2}k_1, t_n + \frac{1}{2}\Delta t), \tag{18}$$

$$y_{n+1} = y_n + \frac{1}{4}k_1 + \frac{3}{4}k_2 + \mathcal{O}\big((\Delta t)^3\big) \tag{19}$$

with corresponding Butcher table

$$\begin{array}{c|cc} 0 & & \\ 2/3 & 2/3 & \\ \hline & 1/4 & 3/4 \end{array}.$$

[6]:
```
# Step 2.a.iv: RK2 Ralston's Method

Butcher_dict['RK2 Ralston'] = (
[[sp.sympify(0)],
[sp.Rational(2,3), sp.Rational(2,3)],
["", sp.Rational(1,4), sp.Rational(3,4)]]
, 2)
```

2.1.5 Step 2.a.v: Kutta's Third-order Method [Back to top]

Kutta's third-order method obtains the solution $y(t)$ at time t_{n+1} from t_n via:

$$k_1 = \Delta t f(y_n, t_n), \tag{20}$$

$$k_2 = \Delta t f(y_n + \frac{1}{2}k_1, t_n + \frac{1}{2}\Delta t), \tag{21}$$

$$k_3 = \Delta t f(y_n - k_1 + 2k_2, t_n + \Delta t) \tag{22}$$

$$y_{n+1} = y_n + \frac{1}{6}k_1 + \frac{2}{3}k_2 + \frac{1}{6}k_3 + \mathcal{O}\big((\Delta t)^4\big) \tag{23}$$

with corresponding Butcher table

$$\begin{array}{c|ccc} 0 & & & \\ 1/2 & 1/2 & & \\ 1 & -1 & 2 & \\ \hline & 1/6 & 2/3 & 1/6 \end{array}.$$

[7]:
```
# Step 2.a.v: Kutta's  Third-order Method

Butcher_dict['RK3'] = (
[[sp.sympify(0)],
[sp.Rational(1,2), sp.Rational(1,2)],
[sp.sympify(1), sp.sympify(-1), sp.sympify(2)],
["", sp.Rational(1,6), sp.Rational(2,3), sp.Rational(1,6)]]
, 3)
```

2.1.6 Step 2.a.vi: RK3 Heun's Method [Back to top]

Heun's third-order method obtains the solution $y(t)$ at time t_{n+1} from t_n via:

$$k_1 = \Delta t f(y_n, t_n), \tag{24}$$

$$k_2 = \Delta t f(y_n + \frac{1}{3}k_1, t_n + \frac{1}{3}\Delta t), \tag{25}$$

$$k_3 = \Delta t f(y_n + \frac{2}{3}k_2, t_n + \frac{2}{3}\Delta t) \tag{26}$$

$$y_{n+1} = y_n + \frac{1}{4}k_1 + \frac{3}{4}k_3 + \mathcal{O}\big((\Delta t)^4\big) \tag{27}$$

with corresponding Butcher table

$$\begin{array}{c|ccc} 0 & & & \\ 1/3 & 1/3 & & \\ 2/3 & 0 & 2/3 & \\ \hline & 1/4 & 0 & 3/4 \end{array}.$$

[8]:
```
# Step 2.a.vi: RK3 Heun's Method

Butcher_dict['RK3 Heun'] = (
[[sp.sympify(0)],
[sp.Rational(1,3), sp.Rational(1,3)],
[sp.Rational(2,3), sp.sympify(0), sp.Rational(2,3)],
["", sp.Rational(1,4), sp.sympify(0), sp.Rational(3,4)]]
, 3)
```

2.1.7 Step 2.a.vii: RK3 Ralton's Method [Back to top]

Ralston's third-order method (see Ralston (1962), obtains the solution $y(t)$ at time t_{n+1} from t_n via:

$$k_1 = \Delta t f(y_n, t_n), \tag{28}$$

$$k_2 = \Delta t f(y_n + \frac{1}{2}k_1, t_n + \frac{1}{2}\Delta t), \tag{29}$$

$$k_3 = \Delta t f(y_n + \frac{3}{4}k_2, t_n + \frac{3}{4}\Delta t) \tag{30}$$

$$y_{n+1} = y_n + \frac{2}{9}k_1 + \frac{1}{3}k_2 + \frac{4}{9}k_3 + \mathcal{O}\big((\Delta t)^4\big) \tag{31}$$

with corresponding Butcher table

0			
1/2	1/2		
3/4	0	3/4	
	2/9	1/3	4/9

.

[9]:
```
# Step 2.a.vii: RK3 Ralton's Method

Butcher_dict['RK3 Ralston'] = (
[[0],
[sp.Rational(1,2), sp.Rational(1,2)],
[sp.Rational(3,4), sp.sympify(0), sp.Rational(3,4)],
["", sp.Rational(2,9), sp.Rational(1,3), sp.Rational(4,9)]]
, 3)
```

2.1.8 Step 2.a.viii: Strong Stability Preserving Runge-Kutta (SSPRK3) Method [Back to top]

The Strong Stability Preserving Runge-Kutta (SSPRK3) method obtains the solution $y(t)$ at time t_{n+1} from t_n via:

$$k_1 = \Delta t f(y_n, t_n), \tag{32}$$
$$k_2 = \Delta t f(y_n + k_1, t_n + \Delta t), \tag{33}$$
$$k_3 = \Delta t f(y_n + \frac{1}{4}k_1 + \frac{1}{4}k_2, t_n + \frac{1}{2}\Delta t) \tag{34}$$
$$y_{n+1} = y_n + \frac{1}{6}k_1 + \frac{1}{6}k_2 + \frac{2}{3}k_3 + \mathcal{O}\big((\Delta t)^4\big) \tag{35}$$

with corresponding Butcher table

0			
1	1		
1/2	1/4	1/4	
	1/6	1/6	2/3

.

[10]:
```
# Step 2.a.viii: Strong Stability Preserving Runge-Kutta (SSPRK3) Method

Butcher_dict['SSPRK3'] = (
[[0],
[sp.sympify(1), sp.sympify(1)],
[sp.Rational(1,2), sp.Rational(1,4), sp.Rational(1,4)],
["", sp.Rational(1,6), sp.Rational(1,6), sp.Rational(2,3)]]
, 3)
```

2.1.9 Step 2.a.ix: Classic RK4 Method [Back to top]

The classic RK4 method obtains the solution $y(t)$ at time t_{n+1} from t_n via:

$$k_1 = \Delta t f(y_n, t_n), \tag{36}$$
$$k_2 = \Delta t f(y_n + \frac{1}{2}k_1, t_n + \frac{\Delta t}{2}), \tag{37}$$
$$k_3 = \Delta t f(y_n + \frac{1}{2}k_2, t_n + \frac{\Delta t}{2}), \tag{38}$$
$$k_4 = \Delta t f(y_n + k_3, t_n + \Delta t), \tag{39}$$
$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) + \mathcal{O}\big((\Delta t)^5\big) \tag{40}$$

with corresponding Butcher table

$$
\begin{array}{c|cccc}
0 & & & & \\
1/2 & 1/2 & & & \\
1/2 & 0 & 1/2 & & \\
1 & 0 & 0 & 1 & \\
\hline
 & 1/6 & 1/3 & 1/3 & 1/6
\end{array}
\,.
$$

[11]:
```
# Step 2.a.vix: Classic RK4 Method

Butcher_dict['RK4'] = (
[[sp.sympify(0)],
[sp.Rational(1,2), sp.Rational(1,2)],
[sp.Rational(1,2), sp.sympify(0), sp.Rational(1,2)],
[sp.sympify(1), sp.sympify(0), sp.sympify(0), sp.sympify(1)],
["", sp.Rational(1,6), sp.Rational(1,3), sp.Rational(1,3), sp.Rational(1,6)]]
, 4)
```

2.1.10 Step 2.a.x: RK5 Dormand-Prince Method [Back to top]

The fifth-order Dormand-Prince (DP) method from the RK5(4) family (see Dormand, J. R.; Prince, P. J. (1980)) Butcher table is:

$$
\begin{array}{c|ccccccc}
0 & & & & & & & \\
\frac{1}{5} & \frac{1}{5} & & & & & & \\
\frac{3}{10} & \frac{3}{40} & \frac{9}{40} & & & & & \\
\frac{4}{5} & \frac{44}{45} & \frac{-56}{15} & \frac{32}{9} & & & & \\
\frac{8}{9} & \frac{19372}{6561} & \frac{-25360}{2187} & \frac{64448}{6561} & \frac{-212}{729} & & & \\
1 & \frac{9017}{3168} & \frac{-355}{33} & \frac{46732}{5247} & \frac{49}{176} & \frac{-5103}{18656} & & \\
1 & \frac{35}{384} & 0 & \frac{500}{1113} & \frac{125}{192} & \frac{-2187}{6784} & \frac{11}{84} & \\
\hline
 & \frac{35}{384} & 0 & \frac{500}{1113} & \frac{125}{192} & \frac{-2187}{6784} & \frac{11}{84} & 0
\end{array}
\,.
$$

[12]:
```
# Step 2.a.x:  RK5 Dormand-Prince Method

Butcher_dict['DP5'] = (
[[0],
[sp.Rational(1,5), sp.Rational(1,5)],
[sp.Rational(3,10),sp.Rational(3,40), sp.Rational(9,40)],
[sp.Rational(4,5), sp.Rational(44,45), sp.Rational(-56,15), sp.Rational(32,9)],
[sp.Rational(8,9), sp.Rational(19372,6561), sp.Rational(-25360,2187), sp.Rational(64448,6561), sp.Rational(-212,729)],
[sp.sympify(1), sp.Rational(9017,3168), sp.Rational(-355,33), sp.Rational(46732,5247), sp.Rational(49,176), sp.Rational(-5103,18656)],
[sp.sympify(1), sp.Rational(35,384), sp.sympify(0), sp.Rational(500,1113), sp.Rational(125,192), sp.Rational(-2187,6784), sp.
↪Rational(11,84)],
```

```
["", sp.Rational(35,384), sp.sympify(0), sp.Rational(500,1113), sp.Rational(125,192), sp.Rational(-2187,6784), sp.Rational(11,84), sp.
↪sympify(0)]]
, 5)
```

2.1.11 Step 2.a.xi: RK5 Dormand-Prince Method Alternative [Back to top]

The fifth-order Dormand-Prince (DP) method from the RK6(5) family (see Dormand, J. R.; Prince, P. J. (1981)) Butcher table is:

$$
\begin{array}{c|ccccccc}
0 & & & & & & & \\
\frac{1}{10} & \frac{1}{10} & & & & & & \\
\frac{2}{9} & \frac{-2}{81} & \frac{20}{81} & & & & & \\
\frac{3}{7} & \frac{615}{1372} & \frac{-270}{343} & \frac{1053}{1372} & & & & \\
\frac{3}{5} & \frac{3243}{5500} & \frac{-54}{55} & \frac{50949}{71500} & \frac{4998}{17875} & & & \\
\frac{4}{5} & \frac{-26492}{37125} & \frac{72}{55} & \frac{2808}{23375} & \frac{-24206}{37125} & \frac{338}{459} & & \\
1 & \frac{5561}{2376} & \frac{-35}{11} & \frac{-24117}{31603} & \frac{899983}{200772} & \frac{-5225}{1836} & \frac{3925}{4056} & \\
\hline
 & \frac{821}{10800} & 0 & \frac{19683}{71825} & \frac{175273}{912600} & \frac{395}{3672} & \frac{785}{2704} & \frac{3}{50}
\end{array}
.
$$

[13]:
```
# Step 2.a.xi:  RK5 Dormand-Prince Method Alternative

Butcher_dict['DP5alt'] = (
[[0],
[sp.Rational(1,10), sp.Rational(1,10)],
[sp.Rational(2,9), sp.Rational(-2, 81), sp.Rational(20, 81)],
[sp.Rational(3,7), sp.Rational(615, 1372), sp.Rational(-270, 343), sp.Rational(1053, 1372)],
[sp.Rational(3,5), sp.Rational(3243, 5500), sp.Rational(-54, 55), sp.Rational(50949, 71500), sp.Rational(4998, 17875)],
[sp.Rational(4, 5), sp.Rational(-26492, 37125), sp.Rational(72, 55), sp.Rational(2808, 23375), sp.Rational(-24206, 37125), sp.
↪Rational(338, 459)],
[sp.sympify(1), sp.Rational(5561, 2376), sp.Rational(-35, 11), sp.Rational(-24117, 31603), sp.Rational(899983, 200772), sp.
↪Rational(-5225, 1836), sp.Rational(3925, 4056)],
["", sp.Rational(821, 10800), sp.sympify(0), sp.Rational(19683, 71825), sp.Rational(175273, 912600), sp.Rational(395, 3672), sp.
↪Rational(785, 2704), sp.Rational(3, 50)]]
, 5)
```

2.1.12 Step 2.a.xii: RK5 Cash-Karp Method [Back to top]

The fifth-order Cash-Karp Method (see J. R. Cash, A. H. Karp. (1980)) Butcher table is:

0						
$\frac{1}{5}$	$\frac{1}{5}$					
$\frac{3}{10}$	$\frac{3}{40}$	$\frac{9}{40}$				
$\frac{3}{5}$	$\frac{3}{10}$	$\frac{-9}{10}$	$\frac{6}{5}$			
1	$\frac{-11}{54}$	$\frac{5}{2}$	$\frac{-70}{27}$	$\frac{35}{27}$		
$\frac{7}{8}$	$\frac{1631}{55296}$	$\frac{175}{512}$	$\frac{575}{13824}$	$\frac{44275}{110592}$	$\frac{253}{4096}$	
	$\frac{37}{378}$	0	$\frac{250}{621}$	$\frac{125}{594}$	0	$\frac{512}{1771}$

[14]:
```
# Step 2.a.xii:  RK5 Cash-Karp Method

Butcher_dict['CK5'] = (
[[0],
[sp.Rational(1,5), sp.Rational(1,5)],
[sp.Rational(3,10),sp.Rational(3,40), sp.Rational(9,40)],
[sp.Rational(3,5), sp.Rational(3,10), sp.Rational(-9,10), sp.Rational(6,5)],
[sp.sympify(1), sp.Rational(-11,54), sp.Rational(5,2), sp.Rational(-70,27), sp.Rational(35,27)],
[sp.Rational(7,8), sp.Rational(1631,55296), sp.Rational(175,512), sp.Rational(575,13824), sp.Rational(44275,110592), sp.
↪Rational(253,4096)],
["",sp.Rational(37,378), sp.sympify(0), sp.Rational(250,621), sp.Rational(125,594), sp.sympify(0), sp.Rational(512,1771)]]
, 5)
```

2.1.13 Step 2.a.xiii: RK6 Dormand-Prince Method [Back to top]

The sixth-order Dormand-Prince method (see Dormand, J. R.; Prince, P. J. (1981)) Butcher Table is:

0								
$\frac{1}{10}$	$\frac{1}{10}$							
$\frac{2}{9}$	$\frac{-2}{81}$	$\frac{20}{81}$						
$\frac{3}{7}$	$\frac{615}{1372}$	$\frac{-270}{343}$	$\frac{1053}{1372}$					
$\frac{3}{5}$	$\frac{3243}{5500}$	$\frac{-54}{55}$	$\frac{50949}{71500}$	$\frac{4998}{17875}$				
$\frac{4}{5}$	$\frac{-26492}{37125}$	$\frac{72}{55}$	$\frac{2808}{23375}$	$\frac{-24206}{37125}$	$\frac{338}{459}$			
1	$\frac{5561}{2376}$	$\frac{-35}{11}$	$\frac{-24117}{31603}$	$\frac{899983}{200772}$	$\frac{-5225}{1836}$	$\frac{3925}{4056}$		
1	$\frac{465467}{266112}$	$\frac{-2945}{1232}$	$\frac{-5610201}{14158144}$	$\frac{10513573}{3212352}$	$\frac{-424325}{205632}$	$\frac{376225}{454272}$	0	
	$\frac{61}{864}$	0	$\frac{98415}{321776}$	$\frac{16807}{146016}$	$\frac{1375}{7344}$	$\frac{1375}{5408}$	$\frac{-37}{1120}$	$\frac{1}{10}$

[15]:
```
# Step 2.a.xiii:  RK6 Dormand-Prince Method

Butcher_dict['DP6'] = (
[[0],
[sp.Rational(1,10), sp.Rational(1,10)],
[sp.Rational(2,9), sp.Rational(-2, 81), sp.Rational(20, 81)],
```

```
[sp.Rational(3,7), sp.Rational(615, 1372), sp.Rational(-270, 343), sp.Rational(1053, 1372)],
[sp.Rational(3,5), sp.Rational(3243, 5500), sp.Rational(-54, 55), sp.Rational(50949, 71500), sp.Rational(4998, 17875)],
[sp.Rational(4, 5), sp.Rational(-26492, 37125), sp.Rational(72, 55), sp.Rational(2808, 23375), sp.Rational(-24206, 37125), sp.
 ↪Rational(338, 459)],
[sp.sympify(1), sp.Rational(5561, 2376), sp.Rational(-35, 11), sp.Rational(-24117, 31603), sp.Rational(899983, 200772), sp.
 ↪Rational(-5225, 1836), sp.Rational(3925, 4056)],
[sp.sympify(1), sp.Rational(465467, 266112), sp.Rational(-2945, 1232), sp.Rational(-5610201, 14158144), sp.Rational(10513573, 3212352),␣
 ↪sp.Rational(-424325, 205632), sp.Rational(376225, 454272), sp.sympify(0)],
["", sp.Rational(61, 864), sp.sympify(0), sp.Rational(98415, 321776), sp.Rational(16807, 146016), sp.Rational(1375, 7344), sp.
 ↪Rational(1375, 5408), sp.Rational(-37, 1120), sp.Rational(1,10)]]
, 6)
```

2.1.14 Step 2.a.xiv: RK6 Luther's Method [Back to top]

Luther's sixth-order method (see H. A. Luther (1968)) Butcher table is:

$$
\begin{array}{c|cccccccc}
0 & & & & & & & & \\
1 & 1 & & & & & & & \\
\frac{1}{2} & \frac{3}{8} & & & & & & & \\
\frac{2}{3} & \frac{8}{27} & \frac{1}{8} & & & & & & \\
\frac{(7-q)}{14} & \frac{(-21+9q)}{392} & \frac{2}{27} & \frac{8}{27} & & & & & \\
\frac{(7+q)}{14} & \frac{(-1155-255q)}{1960} & \frac{(-56+8q)}{392} & \frac{(336-48q)}{392} & \frac{(-63+3q)}{392} & & & & \\
1 & \frac{(330+105q)}{180} & \frac{(-280-40q)}{1960} & \frac{320q}{1960} & \frac{(63+363q)}{1960} & \frac{(2352+392q)}{1960} & & & \\
 & \frac{1}{20} & \frac{2}{3} & \frac{(-200+280q)}{180} & \frac{(126-189q)}{180} & \frac{(-686-126q)}{180} & \frac{(490-70q)}{180} & & \\
 & & 0 & \frac{16}{45} & 0 & \frac{49}{180} & \frac{49}{180} & \frac{1}{20} &
\end{array}
$$

where $q = \sqrt{21}$.

[16]:
```
#  Step 2.a.xiv:  RK6 Luther's Method

q = sp.sqrt(21)
Butcher_dict['L6'] = (
[[0],
[sp.sympify(1), sp.sympify(1)],
[sp.Rational(1,2), sp.Rational(3,8), sp.Rational(1,8)],
[sp.Rational(2,3), sp.Rational(8,27), sp.Rational(2,27), sp.Rational(8,27)],
[(7 - q)/14, (-21 + 9*q)/392, (-56 + 8*q)/392, (336 -48*q)/392, (-63 + 3*q)/392],
[(7 + q)/14, (-1155 - 255*q)/1960, (-280 -  40*q)/1960, (-320*q)/1960, (63 + 363*q)/1960, (2352 + 392*q)/1960],
[sp.sympify(1), ( 330 + 105*q)/180, sp.Rational(2,3), (-200 + 280*q)/180, (126 - 189*q)/180, (-686 - 126*q)/180, (490 -  70*q)/180],
["", sp.Rational(1, 20), sp.sympify(0), sp.Rational(16, 45), sp.sympify(0), sp.Rational(49, 180), sp.Rational(49, 180), sp.Rational(1,␣
 ↪20)]]
, 6)
```

2.1.15 Step 2.a.xv: RK8 Dormand-Prince Method [Back to top]

The eighth-order Dormand-Prince Method (see Dormand, J. R.; Prince, P. J. (1981)) Butcher table is:

0									
$\frac{1}{18}$	$\frac{1}{18}$								
$\frac{1}{12}$	$\frac{1}{48}$	$\frac{1}{16}$							
$\frac{1}{8}$	$\frac{1}{32}$	0	$\frac{3}{32}$						
$\frac{5}{16}$	$\frac{5}{16}$	0	$\frac{-75}{64}$	$\frac{75}{64}$					
$\frac{3}{8}$	$\frac{3}{80}$	0	0	$\frac{3}{16}$	$\frac{3}{20}$				
$\frac{59}{400}$	$\frac{29443841}{614563906}$	0	0	$\frac{77736538}{692538347}$	$\frac{-28693883}{1125000000}$	$\frac{23124283}{1800000000}$			
$\frac{93}{200}$	$\frac{16016141}{946692911}$	0	0	$\frac{61564180}{158732637}$	$\frac{22789713}{633445777}$	$\frac{545815736}{2771057229}$	$\frac{-180193667}{1043307555}$		
$\frac{5490023248}{9719169821}$	$\frac{39632708}{573591083}$	0	0	$\frac{-433636366}{683701615}$	$\frac{-421739975}{2616292301}$	$\frac{100302831}{723423059}$	$\frac{790204164}{839813087}$	$\frac{800635310}{3783071287}$	
$\frac{13}{20}$	$\frac{246121993}{1340847787}$	0	0	$\frac{-37695042795}{15268766246}$	$\frac{-309121744}{1061227803}$	$\frac{-12992083}{490766935}$	$\frac{6005943493}{2108947869}$	$\frac{393006217}{1396673457}$	$\frac{123872331}{1001029789}$
$\frac{1201146811}{1299019798}$	$\frac{-1028468189}{846180014}$	0	0	$\frac{8478235783}{508512852}$	$\frac{1311729495}{1432422823}$	$\frac{-10304129995}{1701304382}$	$\frac{-48777925059}{3047939560}$	$\frac{15336726248}{1032824649}$	$\frac{-45442868181}{3398467696}$
$\frac{3065993473}{597172653}$	$\frac{185892177}{718116043}$	0	0	$\frac{-3185094517}{667107341}$	$\frac{-477755414}{1098053517}$	$\frac{-703635378}{230739211}$	$\frac{5731566787}{1027545527}$	$\frac{5232866602}{850066563}$	$\frac{-4093664535}{808688257}$
1	$\frac{65686358}{487910083}$								
$\frac{3962137247}{1805957418}$	$\frac{403863854}{491063109}$	0	0	$\frac{-5068492393}{434740067}$	$\frac{-411421997}{543043805}$	$\frac{652783627}{914296604}$	$\frac{11173962825}{925320556}$	$\frac{-13158990841}{6184727034}$	$\frac{3936647629}{1978049680}$
1	$\frac{248638103}{1413531060}$	0							
$\frac{-160528059}{685178525}$	$\frac{14005451}{335480064}$	0	0	0	0	$\frac{-59238493}{1068277825}$	$\frac{181606767}{758867731}$	$\frac{561292985}{797845732}$	$\frac{-1041891430}{1371343529}$
$\frac{760417239}{1151165299}$	$\frac{118820643}{751138087}$	$\frac{-528747749}{2220607170}$	$\frac{1}{4}$						

[17]:
```
# Step 2.a.xv:  RK8 Dormand-Prince Method

Butcher_dict['DP8']=(
[[0],
[sp.Rational(1, 18), sp.Rational(1, 18)],
[sp.Rational(1, 12), sp.Rational(1, 48), sp.Rational(1, 16)],
[sp.Rational(1, 8), sp.Rational(1, 32), sp.sympify(0), sp.Rational(3, 32)],
[sp.Rational(5, 16), sp.Rational(5, 16), sp.sympify(0), sp.Rational(-75, 64), sp.Rational(75, 64)],
[sp.Rational(3, 8), sp.Rational(3, 80), sp.sympify(0), sp.sympify(0), sp.Rational(3, 16), sp.Rational(3, 20)],
[sp.Rational(59, 400), sp.Rational(29443841, 614563906), sp.sympify(0), sp.sympify(0), sp.Rational(77736538, 692538347), sp.
 ↪Rational(-28693883, 1125000000), sp.Rational(23124283, 1800000000)],
[sp.Rational(93, 200), sp.Rational(16016141, 946692911), sp.sympify(0), sp.sympify(0), sp.Rational(61564180, 158732637), sp.
 ↪Rational(22789713, 633445777), sp.Rational(545815736, 2771057229), sp.Rational(-180193667, 1043307555)],
[sp.Rational(5490023248, 9719169821), sp.Rational(39632708, 573591083), sp.sympify(0), sp.sympify(0), sp.Rational(-433636366,␣
 ↪683701615), sp.Rational(-421739975, 2616292301), sp.Rational(100302831, 723423059), sp.Rational(790204164, 839813087), sp.
 ↪Rational(800635310, 3783071287)],
```

```
[sp.Rational(13, 20), sp.Rational(246121993, 1340847787), sp.sympify(0), sp.sympify(0), sp.Rational(-37695042795, 15268766246), sp.
↪Rational(-309121744, 1061227803), sp.Rational(-12992083, 490766935), sp.Rational(6005943493, 2108947869), sp.Rational(393006217,␣
↪1396673457), sp.Rational(123872331, 1001029789)],
[sp.Rational(1201146811, 1299019798), sp.Rational(-1028468189, 846180014), sp.sympify(0), sp.sympify(0), sp.Rational(8478235783,␣
↪508512852), sp.Rational(1311729495, 1432422823), sp.Rational(-10304129995, 1701304382), sp.Rational(-48777925059, 3047939560), sp.
↪Rational(15336726248, 1032824649), sp.Rational(-45442868181, 3398467696), sp.Rational(3065993473, 597172653)],
[sp.sympify(1), sp.Rational(185892177, 718116043), sp.sympify(0), sp.sympify(0), sp.Rational(-3185094517, 667107341), sp.
↪Rational(-477755414, 1098053517), sp.Rational(-703635378, 230739211), sp.Rational(5731566787, 1027545527), sp.Rational(5232866602,␣
↪850066563), sp.Rational(-4093664535, 808688257), sp.Rational(3962137247, 1805957418), sp.Rational(65686358, 487910083)],
[sp.sympify(1), sp.Rational(403863854, 491063109), sp.sympify(0), sp.sympify(0), sp.Rational(-5068492393, 434740067), sp.
↪Rational(-411421997, 543043805), sp.Rational(652783627, 914296604), sp.Rational(11173962825, 925320556), sp.Rational(-13158990841,␣
↪6184727034), sp.Rational(3936647629, 1978049680), sp.Rational(-160528059, 685178525), sp.Rational(248638103, 1413531060), sp.
↪sympify(0)],
["", sp.Rational(14005451, 335480064), sp.sympify(0), sp.sympify(0), sp.sympify(0), sp.sympify(0), sp.Rational(-59238493, 1068277825),␣
↪sp.Rational(181606767, 758867731), sp.Rational(561292985, 797845732), sp.Rational(-1041891430, 1371343529), sp.Rational(760417239,␣
↪1151165299), sp.Rational(118820643, 751138087), sp.Rational(-528747749, 2220607170), sp.Rational(1, 4)]]
, 8)
```

[18]:
```
Butcher_dict['AHE'] = (
[[sp.sympify(0)],
[sp.sympify(1), sp.sympify(1)],
["", sp.Rational(1,2), sp.Rational(1,2)],
["", sp.sympify(1), sp.sympify(0)]]
, 2)
```

3 Step 3: Code validation against MoLtimestepping.RK_Butcher_Table_Dictionary NRPy+ module [Back to top]

As a code validation check, we verify agreement in the dictionary of Butcher tables between 1. this tutorial and 2. the NRPy+ MoLtimestepping.RK_Butcher_Table_Dictionary module.

We analyze all key/value entries in the dictionary for consistency.

[19]:
```
# Step 3: Code validation against MoLtimestepping.RK_Butcher_Table_Dictionary NRPy+ module
import sys  # Standard Python module for multiplatform OS-level functions
from MoLtimestepping.RK_Butcher_Table_Dictionary import Butcher_dict as B_dict
valid = True
for key, value in Butcher_dict.items():
    if Butcher_dict[key] != B_dict[key]:
        valid = False
        print(key)
```

```
if valid == True and len(Butcher_dict.items()) == len(B_dict.items()):
    print("The dictionaries match!")
else:
    print("ERROR: Dictionaries don't match!")
    sys.exit(1)
```

```
The dictionaries match!
```

4 Step 4: The Family of Adaptive Runge-Kutta-Like Schemes (Butcher Tables) [Back to top]

In the previous parts of this notebook, we worked exclusively with Explicit Runge-Kutta-Like Schemes. There are a second category of schemes that are useful, however, and those are the Adaptive Runge-Kutta-Like Schemes. These methods, while technically still explicit in function, nonetheless need to be implemented differently. They have an extra row at the bottom of their butcher tables which define a second method of a different order. Adaptive Runge-Kutta-Like methods will calculate the predicted values of both results and use the difference between them to estimate the error in a highly efficient manner. Details can be found on the Runge-Kutta wikipedia page.

As a reminder, the general result of any Runge-Kutta-Like scheme is

$$y_{n+1} = y_n + \sum_{i=1}^{s} b_i k_i.$$

An Adaptive method will also calculate the following

$$y^*_{n+1} = y_n + \sum_{i=1}^{s} b^*_i k_i$$

Where the b^* coefficients are stored in the butcher table on an extra row, like so:

$$\begin{array}{c|ccccc} 0 & & & & & \\ c_2 & a_{21} & & & & \\ c_3 & a_{31} & a_{32} & & & \\ \vdots & \vdots & & \ddots & & \\ c_s & a_{s_1} & a_{s2} & \cdots & a_{s,s-1} & \\ \hline & b_1 & b_2 & \cdots & b_{s-1} & b_s \\ & b^*_1 & b^*_2 & \cdots & b^*_{s-1} & b^*_s \end{array}$$

From this method the error estimate is calcualted as $ e_{n+1} = y_{n+1} - y^*_{n+1} $. The exact use of this error is implementation-dependent, but in general if the error exceeds a provided threshhold the scheme will discard the step and try again at a smaller timestep, and if the error is below a provided threshhold the scheme will increase the step size (but in general it will not discard the data, it already calculated it, why waste it?).

4.1 Step 4.a: Generating a Dictionary of Butcher Tables for Explicit Runge Kutta Techniques [Back to top]

As before, we can store all of the Butcher tables in Python's **Dictionary** format using the curly brackets {} and 'key':value pairs. The 'key' will be the *name* of the Runge Kutta method and the value will be the Butcher table itself stored as a list of lists. The convergence order for each Runge Kutta method is also stored. We will construct the dictionary `Butcher_dict` one Butcher table at a time in the following sections.

Adaptive methods are all prefaced with an A in their name, for clarity's sake.

4.1.1 Step 4.a.i: Adaptive Heun-Euler Method [Back to top]

The Heun-Euler method is a combination of Heun's second-order method and the first-order Euler method, creating the simplest Adaptive method.

$$\begin{array}{c|cc} 0 & & \\ 1 & 1 & \\ \hline & 1/2 & 1/2 \\ & 1 & 0 \end{array}$$

[24]:
```
Butcher_dict['AHE'] = (
[[sp.sympify(0)],
[sp.sympify(1), sp.sympify(1)],
["", sp.Rational(1,2), sp.Rational(1,2)],
["", sp.sympify(1), sp.sympify(0)]]
, 1)
```

4.1.2 Step 4.a.ii: Adaptive Bogacki-Shampine Method [Back to top]

The Bogacki-Shampine Method is a second order method with a third order error estimator.

$$\begin{array}{c|ccc} 0 & & & \\ 1/2 & 1/2 & & \\ 3/4 & 0 & 3/4 & \\ & & & \\ 1 & 2/9 & 1/3 & \\ 4/9 & & & \\ \hline & 2/9 & 1/3 & \\ 4/9 & 0 & & \\ & 7/24 & 1/4 & \\ 1/3 & 1/8 & & \end{array}$$

[25]:
```
Butcher_dict['ABS'] = (
[[sp.sympify(0)],
[sp.Rational(1,2), sp.Rational(1,2)],
[sp.Rational(3,4), sp.sympify(0), sp.Rational(3,4)],
[sp.sympify(1), sp.Rational(2,9), sp.Rational(1,3), sp.Rational(4,9)],
["", sp.Rational(2,9), sp.Rational(1,3), sp.Rational(4,9), sp.sympify(0)],
["", sp.Rational(7,24), sp.Rational(1,4), sp.Rational(1,3), sp.Rational(1,8)]]
, 3)
```

4.1.3 Step 4.a.iii: Adaptive Runge-Kutta-Fehlberg [Back to top]

The Runge-Kutta-Fehlberg Method is a fourth order method with a fifth order corrective.

0			
1/4	1/4		
3/8	3/32	9/32	
12/13	1932/2197	−7200/2197	
7296/2197			
1	439/216	−8	
3680/513	−845/4104		
1/2	−8/27	2	
−3544/2565	1859/4104	−11/40	
	16/135	0	
6656/12825	28561/56430	−9/50	
2/55			
	25/216	0	
1048/2565	2197/4104	−1/5	
0			

[27]:
```
Butcher_dict['ARKF'] = (
[[sp.sympify(0)],
[sp.Rational(1,4), sp.Rational(1,4)],
[sp.Rational(3,8), sp.Rational(3,32), sp.Rational(9,32)],
[sp.Rational(12,13), sp.Rational(1932,2197), sp.Rational(-7200,2197), sp.Rational(7296,2197)],
[sp.sympify(1), sp.Rational(439,216), sp.sympify(-8), sp.Rational(3680,513), sp.Rational(-845,4104)],
[sp.Rational(1,2), sp.Rational(-8,27), sp.sympify(2), sp.Rational(-3544,2565), sp.Rational(1859.4104), sp.Rational(-11,40)],

["", sp.Rational(16,135), sp.sympify(0), sp.Rational(6656,12825), sp.Rational(28561,56430), sp.Rational(-9,50), sp.Rational(2,55)],
["", sp.Rational(25,216), sp.sympify(0), sp.Rational(1048,2565), sp.Rational(2197,4104), sp.Rational(-1,5), sp.sympify(0)]]
, 4)
```

5 Step 5: Output this notebook to LaTeX-formatted PDF file [Back to top]

The following code cell converts this Jupyter notebook into a proper, clickable LaTeX-formatted PDF file. After the cell is successfully run, the generated PDF may be found in the root NRPy+ tutorial directory, with filename Tutorial-RK_Butcher_Table_Dictionary.pdf. (Note that clicking on this link may not work; you may need to open the PDF file through another means.)

[22]:
```
import cmdline_helper as cmd    # NRPy+: Multi-platform Python command-line interface
cmd.output_Jupyter_notebook_to_LaTeXed_PDF("Tutorial-RK_Butcher_Table_Dictionary")
```

```
Created Tutorial-RK_Butcher_Table_Dictionary.tex, and compiled LaTeX file
    to PDF file Tutorial-RK_Butcher_Table_Dictionary.pdf
```